INCH-POUND

MIL-C-23842B(SH)
14 July 1993

SUPERSEDING
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22 June 1982
(See 6.10)

MILITARY SPECIFICATION

CONVEYOR, VERTICAL, TRAY TYPE, FOR 3000 POUND PALLETIZED LOADS (SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers powered tray type vertical conveyors used to move palletized loads up to 3000 pounds aboard Naval ships. Only one classification of conveyor is covered by this specification.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: 03Q42 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3910

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FF-B-185 - Bearings, Roller, Cylindrical; and Bearings Roller, Self-aligning.

TT-C-490 - Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.

TT-P-645 - Primer, Paint, Zinc-Molybdate, Alkyd Type.

TT-P-1757 - Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.

MIL-C-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices.

MIL-M-3184 - Machinery: Deck and Vehicle Mounted with Associated Equipment and Provisioned (Repair Parts) Items; Packaging of.

MIL-M-7793 - Meter, Time Totalizing.

MIL-M-7793/8 - Meter, Time Totalizing, Miniature, Digital, 115 Volts, 60 or 400 Hz.

MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.

MIL-P-15024/5 - Plates, Identification.

MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).

MIL-C-15306 - Call-Signal Stations (Type IC/D) (Hand Crank Magneto and Call-Signal).

MIL-T-15514 - Telephone Equipment, Sound Powered Handsets, Headset-Chest Sets, and Headset-Microphones, Types H-200/U, H-201/U, H-202/U, H-203/U, and H-204/U.

MIL-B-16392 - Brakes, Magnet, Naval Shipboard.

MIL-M-17060 - Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.

MIL-C-24707 - Castings, Ferrous, General Specification for.

MIL-C-24707/5 - Castings, Ductile Iron and Austenitic Ductile Iron.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

FED-STD-H28/2 - Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads-UN and UNR Thread Forms.

MILITARY

MIL-STD-278 - Welding and Casting Standard.

MIL-STD-471 - Maintainability Verification/Demonstration/Evaluation.

MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.

HANDBOOKS

MILITARY

MIL-HDBK-267 - Guide For Selection of Lubricants and Hydraulic Fluids For Use In Shipboard Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

American Gear Manufacturers Association Gear Handbook

(Application for copies should be addressed to the American Gear Manufacturers Association, 1500 King St., Suite 201, Alexandria, VA 22314.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B29.1 - Precision Power Transmission Roller Chains, Attachments, and Sprockets. (DOD adopted)

S1.4 - Specification for Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36/A 36M - Standard Specification for Structural Steel. (DOD adopted)

A 47 - Standard Specification for Ferritic Malleable Iron Castings. (DOD adopted)

A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip. (DOD adopted)

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be free from any defects and imperfections that might affect the serviceability or appearance of the finished product. Materials shall be in compliance with the requirements specified herein. Materials not definitely specified herein shall be of a quality to meet the performance requirements specified herein. Where materials are not identified the following order of preference shall apply:

(a) Industry or technical society specification or standard.
(b) Federal specification.
(c) Military specification.
(d) Manufacturer's specification or standard.

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Where materials other than those covered by the material references are to be used, the drawings shall show the complete chemical and physical properties of the material. Components shall be compatible for use with lubricants as specified in 3.2.3.

3.1.1 Cast iron. Cast iron in any form shall not be used except where permitted by referenced specifications. The use of malleable cast iron is limited to ASTM A 47. Cast nodular graphitic iron in accordance with MIL-C-24707 and MIL-C-24707/5 may be used for machinery equipment components.

3.1.2 Cam track. Cam track, guides for the tray guide arm rollers, shall be corrosion resistant steel in accordance with ASTM A 167, type 304 and shall be of the uninterrupted cam track design.

3.1.3 Roller chains. Roller chains, their attachments, and all hardware shall be in accordance with ANSI B29.1.

3.1.4 Chain sprockets. Chain sprockets shall be in accordance with ANSI B29.1.

3.1.5 Other components. Component parts of the conveyor shall be constructed or selected for minimum weight consistent with stress requirements herein and shall meet design and operational requirements.

3.1.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Construction. Each conveyor shall be a complete unit, including trunk, cam tracks, tray guides, liners, trays, overload devices, sprocket and sprocket assemblies, chains, chain tensioner, controller, pushbutton stations, interlocks, photo cells, and associated mechanical and electrical drive equipment. The configuration of the conveyor shall be in accordance with figure 1 (see 6.3, 6.5, and appendix).

3.2.1 Structure. Where practicable, welded steel construction shall be used in lieu of castings.

3.2.2 Fasteners. Keys, collars, nuts, bolts, studs, setscrews, and other parts that tend to work loose shall be fastened in a positive manner using cotter pins, safety wires, or other suitable methods. Threaded parts shall have threads in accordance with FED-STD-H28 and FED-STD-H28/2.

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3.2.3 Lubrication. Means for the lubrication of all metallic surfaces in sliding or rolling contact shall be provided. Lubrication fittings shall be accessible without the disassembly of any components. Cam tracks shall not require lubrication. Lubricants shall be in accordance with MIL-HDBK-267.

3.2.4 Bearings. Rotating parts shall be mounted on ball or roller bearings to suit the intended use, in accordance with FF-B-171 and FF-B-185. Bearings shall be selected for 10,000 hours minimum operating life under rated conveyor operating loads and as recommended by the bearing manufacturer, with load ratings, including applicable safety factors, greater than the maximum bearing loads under extreme operating conditions specified herein (see 6.3 and 6.7). Shaft bearings shall be sealed to keep out dirt and moisture.

3.2.5 Liners. Liners shall be provided to prevent jamming due to a pallet shifting in any direction on a moving tray during transit and to afford maximum protection to operating personnel. Liners shall have a smooth configuration to prevent pallets from being snagged. Conveyor liners shall be mounted and fastened to permit removal. The rear liner shall be built to withstand impact from palletized load, without being deformed, when load is being positioned on tray by forklift truck.

3.2.6 Carrier chains. Adjustment shall be provided for tension of the carrier chains. Adjustment shall allow for removal of the tail sprocket without parting the chain. Provision shall be made for disengagement of sections of the carrier chains between trays to permit repairs.

3.2.7 Tracks. Cam tracks shall provide positive guidance for tray travel without binding of tray motion and shall provide a smooth transition of each tray as it travels from its horizontal (load carrying) position to its vertical (idle return) position. Vertical track sections shall be replaceable by bolting or other non-welded means. Tail cam track panels shall have fixed (dead) center chain sprocket arrangements.

3.2.8 Base cleanout access. A cleanout door shall be provided at the base of the structure to provide means of keeping the conveyor interior clear of debris. Access opening shall be not less than 18 by 24 inches.

3.2.9 Frame. The conveyor frame shall be the unitized trunk type, and shall be constructed of steel in accordance with ASTM A 36/A 36M. The complete conveyor assembly or each section of conveyor length shall be provided with lifting attachments to permit handling of unit in a vertical attitude. The trunk shall withstand pressures required to meet the tightness requirements (i.e air tight, oil tight, watertight, etc.) of the ship in which conveyor will be installed (see 6.2).

3.2.10 Configuration. Unless otherwise specified (see 6.2), the conveyor shall have a configuration as shown on figure 1. The conveyor shall serve decks as specified (see 6.2).

3.2.10.1 Trays. The number of trays shall be determined by the nominal tray spacing (see 6.2) and the number of decks penetrated by the conveyor.

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3.2.10.2 Machinery. The machinery arrangement shall be for right-hand or left-hand drive as specified (see 6.2.).

3.2.10.3 Sections. Conveyors greater than 50 feet overall length may be made up of two mating sections approximately equal in length. The section joint shall equal the maximum strength of the section.

3.2.10.4 Welded fabrication. Welded fabrication and inspection shall be in accordance with MIL-STD-278.

3.2.11 Ship motion.

3.2.11.1 Conveyor alignment. Design and installation of the trunk and conveyor shall result in a structure which shall maintain alignment and shall operate in accordance with the requirements of this specification after the ship has been subjected to a 30-degree roll, a 4-degree pitch, or combination of the two.

3.2.11.2 Conveyor operation. The conveyor shall operate without adverse effect with a conveyor rated capacity load at rated speed under any combination of the following ship motion combinations:

(a) Roll - From 15 degrees port to 15 degrees starboard and return in a full period of 9 seconds.

(b) Pitch - From 4 degrees up to 4 degrees down and return in a full period of 6 seconds.

The conveyor shall also operate without adverse effect with a conveyor rated capacity load at rated speed when the ship is permanently listed 15 degrees to one side and trimmed 4 degrees down by the bow or by the stern.

3.2.12 High-impact shock. When specified (see 6.2), the conveyor drive machinery shall comply with the requirements for grade B equipment when tested as specified. Certification of compliance with the test requirements for high-impact shock, from a Government approved test facility is acceptable in lieu of the shipboard shock test (see 4.6.8).

(a) Test specification - MIL-S-901.
(b) Shockproofness required - grade B.
(c) Equipment classification - deck mounted.
(d) Class of equipment - class I.
(e) Class of test - medium weight test.
(f) Type of test - type B.

3.3 Design stress. For rated conditions, the combined stress in any structural or operating part of the conveyor shall be not greater than 35 percent of the yield strength of the material used. The combined stress in any structural or operating component of the conveyor shall be not greater than 70 percent of the yield strength of its material when the conveyor is subjected to a

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static load equal to 200 percent of tray capacity multiplied by the number of trays in the horizontal position (see 4.6.1 for static overload test), or when the conveyor is subjected to a tray pallet jam condition and the load on a single tray (due to motor torque) is equal to 115 percent of the conveyor rated capacity, whichever produces the greater stress.

3.3.1 Calculation. In design stress calculations, the stress relationships shown in table I shall apply.

TABLE I. *Design stress calculations.*

Stress (lb/in^2)	Percent of tensile yield stress
Direct shear	60
Torsional shear	65
Compression (bearing)	160

3.3.2 Drive and tray carrier chain. Drive and tray carrier chain shall be in accordance with ANSI B29.1, and be of roller chain construction with a safety factor of not less than 10 based on the ratio of the ultimate strength to the working load (conveyor rated capacity load plus dead load). A factor of safety of not less than 12 shall be used for tray carrier chains of conveyors over 50 feet long (center-to-center distance between head and tail sprockets).

3.4. Structure. The conveyor shall consist of a compact structure vertically supported at the base and laterally at the intermediate deck levels and laterally at the top. Sufficient clearance shall be provided within the body of the conveyor to allow safe and easy unloading and loading of palletized cargo (see 3.4.2).

3.4.1 Trays. The trays shall be supported on two sides by the carrier chains and shall be guided throughout the entire vertical travel by guide arms with rollers attached to the sides of the tray and ride in tracks mounted in the conveyor. Guide arms shall be keyed to tray mounting shafts. The tray shall ride in normal horizontal position at the front of the conveyor and shall be rotated to a vertical position for the idle return portion of the conveyor cycle. The rotation and translation of the tray around the chain sprockets shall be accomplished by the arrangement of the tracks and the camming action of the tray guide arms tracking in the tracks. The trays shall be positively guided at all times. The tray size shall be 45 inches deep by 60 inches wide with a continuous surface.

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3.4.2 Dimensions. The conveyor shall handle palletized cargo 45 inches deep, by 54 inches wide, by 56 inches high on a standard 40 inch by 48-inch pallet. The load area within the conveyor shall have dimensions of not less than 66 inches wide by 53 inches deep. The conveyor cross section maximum dimensions shall be not greater than 96 inches wide by 74 inches deep. For side mounted drive assembly, conveyor width shall be not greater than 121 inches.

3.4.3 Trunk doors and openings. A trunk door shall be provided at each load station to secure the trunk opening. These doors shall be designed to the same strength requirements as the trunk and shall meet the tightness requirements of the trunk as specified (see 6.2). Unless otherwise specified, (see 6.2), the trunk doors shall have a clear opening of 60 inches high by 66 inches wide. The doors shall positioned so that the trays align with the sills at all levels served simultaneously.

3.5 Performance. The conveyor shall be capable of transferring 3000 pound pallet loads between the deck with the master control station and any other of the decks served and shall move the 3000 pound pallet loads between any of these decks in response to operator commands. Operation of the conveyor shall be by manual control and shall provide for automatic stop at each level. Automatic stop shall be accomplished by the tripping of limit travel switches (in the UP and DOWN circuits) by a cam segment mounted on the conveyor head shaft drive sprocket. Electrical controls shall be provided to operate the conveyor in the UP and DOWN direction and to STOP the conveyor. For each cycle of travel of trays, the magnetic clutch shall be engaged and the brake released.

3.5.1 Tray capacity, conveyor rated capacity, and speed. Unless otherwise specified (see 6.2), the tray capacity of each tray shall be 3000 pounds; the conveyor rated capacity shall be the tray capacity multiplied by the number of trays in the horizontal position, minus one tray; tray speed shall be 36 to 38 feet per minute.

3.6 Conveyor drive assembly. The conveyor drive assembly shall be a direct drive from an electric motor to a magnetic clutch, then to a non-overhauling speed reducer, and then a dual chain and sprocket drive. A solenoid operated brake shall be mounted on the extension of the input shaft of the speed reducer. Dowels, lifting eyes, and forcing bolts shall be provided as necessary to facilitate the installation, maintenance, and repair of the conveyor drive assembly.

3.6.1 Drive machinery. Drive machinery shall consist of the electric motor and its controller, magnetic clutch, and speed reducer with brake. Drive machinery shall operate at the conveyor rated capacity continuously with 90 starts and stops per hour in either direction (see 4.6.3) and at 150 percent of the conveyor rated capacity for one cycle (see 4.6.2). Starting and stopping of the tray motion shall be by means of the magnetic clutch and the solenoid operated friction brake.

3.6.1.1 Electric drive motor. The electric motor shall be in accordance with MIL-M-17060 and the following:

(a) Service - C.

(b) Ambient temperature - 50 degrees Celsius (°C).

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(c) Voltage and electrical characteristics - 440-volt (V), 3-phase, 60-hertz (Hz).

(d) Bearings - ball bearing.

(e) Enclosure - dripproof (see 6.2).

(f) Horsepower - as required to operate conveyor continuously (90 starts and stops per hour) at conveyor rated capacity (see 3.5.1) and at rated speed or to operate the conveyor for one cycle (see 4.6.9.1) a 150 percent of conveyor rated capacity, whichever is greater.

(g) Speed - constant.

(h) Duty - continuous.

(i) Insulation - class B.

(j) Type - squirrel cage induction, design B.

(k) Temperature rise - class B insulation limits.

3.6.1.2 Electric motor controller. The electric motor controller shall be in accordance with MIL-C-2212 and the following:

(a) Voltage and electrical characteristics - 440-V, 3-phase, 60-Hz.

(b) Operation - magnetic.

(c) Type - across-the-line.

(d) Performance - semi-automatic.

(e) Protection - low voltage protection and motor thermal overload relays.

(f) Function - motor starting and reversing.

(g) Enclosure - dripproof (see 6.2).

(h) Ambient temperature - 50°C.

(i) Direct current (dc) power supply for the magnetic clutch and the communications system shall be mounted in the controller enclosure.

(j) Other features:

(1) Control circuit power ON-OFF switch - two position lock key type with key

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removable at each position.

(2) Indicator light (red) wired to power ON-OFF switch.

(3) The control circuit power ON-OFF switch and indicator light shall be installed in the face panel of the controller. Operating control circuits shall be 110-V. The motor controller shall be located external to the trunk in the immediate vicinity of the master control station.

(4) The electrical overload device shall be connected to the motor control circuitry. A meter readout shall be provided and installed as specified in 3.8.3.

3.6.1.3 Clutch. The clutch shall be commercial marine in accordance with the following:

(a) Type - magnetic, disc.

(b) Voltage - direct current which shall be obtained through the use of a transformer/rectifier, and shall be fed through the collector rings mounted on hub. The rectifier shall be mounted in controller enclosure.

(c) Duty - continuous (capable of 90 operations per hour with a conveyor rated capacity load without over-heating and with minimum wear). Clutch torque rating shall provide a safety factor of not less than 3 based on the full load torque requirement (see 6.3 and 6.7). The magnetic clutch shall have a variable range to cover the operational and test requirements specified and shall provide over-torque protection for the drive motor in the event of a jam or excessive overload equal to 150 percent of the conveyor rated capacity load (see 4.6.2 for dynamic overload test).

3.6.1.4 Speed reducer. The speed reducer shall be of the non-overhauling type, that is, the input shaft of the speed reducer shall not be driven by external forces applied to the output shaft (see 4.6.4). Gears shall be in accordance with AGMA Gear Handbook, gear quality number as specified for conveyor equipment. Gears shall be fully enclosed, operate in a bath of oil and be mounted on anti-friction bearings. Gear casings shall be provided with access plates to facilitate inspection, repair, cleaning, and removal of gears. A sight gauge indicating the minimum and maximum oil levels in the case and showing the entire range between these levels shall be provided. There shall be access openings for lubricant fill, drain, and breather which may be easily opened and closed.

3.6.1.5 Brake. The brake shall be in accordance with MIL-B-16392 and the following:

(a) Type - disc.

(b) Ambient temperature - 50 °C.

(c) Voltage and electrical characteristics - 440- V, single-phase, 60- Hz.

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(d) Enclosure - dripproof (see 6.2).

(e) Duty - continuous (90 starts and stops per hour).

(f) Mounting - reducer mounted.

(g) Torque - as required to meet the 150 percent dynamic overload test.

3.6.1.6 Machinery bedplate. The drive machinery shall be doweled to a bedplate. The bedplate shall be of adequate strength to maintain an alignment without aid from the deck foundation. It shall be welded steel construction and furnished with flat underside bearing surface and with spot-faced upper side. The thickness of the bedplate flanges shall be not less than the diameter of the hold-down bolts used. Foundation bolt holes shall be drilled 1/10 inch undersize for reaming at installation to fit body-bound bolts.

3.6.2 Chain and sprocket assembly. A dual chain and chain sprocket from the output shaft of the speed reducer shall power the conveyor jackshaft. A chain and chain sprocket from the output sprocket of the jackshaft shall power each end of the conveyor headshaft.

3.6.2.1 Chain tensioners. Chain tensioning adjustable take-up idler sprockets shall be provided for adjustment of the chain tension. Adjustable take-up idler sprockets shall be located near the small sprocket and on the outside of the drive chain.

3.6.2.2 Chain sprocket assembly. Chain sprockets shall be keyed to preclude shifting out of position on the shaft.

3.6.2.3 Shafts. Rotating shafts shall be supported in anti-friction bearings. Head shafts shall be of forged or welded construction. If welded construction is used, welds shall be X-rayed and shall comply with requirements for welded fabrication in accordance with MIL-STD-278.

3.7 Controls. The conveyor when started shall cycle (distance between adjacent trays) and stop when the next available tray is aligned with the load-unload position at the load station. The conveyor shall either stop or fail to start under the following conditions:

(a) Control circuit open - photo-electric beam interrupted due to:

(1) Load out of position on tray.

(2) Tray load carried past the top or bottom load station.

(b) Actuation of the electrical overload device, STOP push-pull switch, or EMERGENCY STOP pushbutton.

For each cycle of travel of trays, the magnetic clutch shall be engaged and the brake released. No control station shall be able to override the function of any safety device.

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3.7.1 Control stations and devices. A control station, in accordance with MIL-C-2212, shall be provided at each conveyor load station and in the machinery space. Unless otherwise specified (see 6.2), each control station shall be located adjacent to the trunk opening, with a clear and unobstructed view of the opening, to facilitate operating the conveyor. Control stations shall be configured as shown on figure 2 and shall be equipped with a lockable security cover to prevent unauthorized conveyor operation. The lockable security cover in the closed position shall not obstruct the station indicator lights and the STOP-ON push-pull switch. The control station in the machinery space shall be mounted near the machinery and shall be within reach and accessible to personnel performing maintenance functions. At each control station, there shall be a sound powered telephone station (see 3.15) and an independent "EMERGENCY STOP" switch (see 3.8.6) located adjacent to the control station. The control circuitry shall be arranged so that the conveyor can be stopped from any control station; but conveyor motion initiated only from the master control station. The control stations shall consist of the following:

(a) One master control station, installed on the specified deck (see 6.2), shall include the following equipment:

(1) A two position selector switch (UP and DOWN) for direction of travel.

(2) A STOP-ON push-pull switch: the ON position shall interlock with the MOTOR START pushbutton switch; the STOP position to de-energize the motor circuit, the magnetic clutch, and set the brake.

(3) A CYCLE pushbutton switch to energize the magnetic clutch, release the brake, to initiate conveyor motion when the motor is running.

(4) LOAD-OUT-OF-POSITION (red) indicator lights for all control stations.

(5) A (green) indicator light for station POWER-ON.

(6) A (white) indicator light for MOTOR-RUNNING.

(7) READY-TO-RUN (white) indicator lights to indicate the status of each remote control stations (see figure 2).

(8) A MOTOR START pushbutton switch to energize the electric drive motor.

(9) An EMERGENCY RUN pushbutton switch to by-pass the electric drive motor thermal overload and low voltage protection relays.

(b) Each control station, other than master control station, shall be a remote control station and shall include the following equipment:

(1) A STOP-ON push-pull switch, the ON position shall interlock with the control circuit of the ON position of the STOP-ON push-pull switch at the master control

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station; the STOP position shall de-energize the motor circuit, the magnetic clutch, and set the brake.

(2) LOAD-OUT-OF-POSITION (red) indicator light.

(3) READY-TO-RUN pushbutton switch. Provide door switches, as required, to bypass the READY-TO-RUN switch for load stations that are closed during conveyor operation.

(4) READY-TO-RUN (white) indicator light.

(5) A (green) indictor light for station POWER-ON.

3.7.1.1 Requirements. The control stations, enclosures, switches, and indicator light shall be in accordance with MIL-C-2212 and the following:

(a) Selector switch (UP and DOWN), maintained-contact type.

(b) STOP-ON push-pull switch, maintained-contact type. The STOP-ON switching function shall be combined into a push (STOP) and a pull (ON) type single switch element and shall have a red mushroom type head for ease of tripping. The STOP-ON switch shall include a visible indicator light (red) when the switch is in the STOP position.

(c) The CYCLE, MOTOR START, and EMERGENCY RUN pushbutton switches shall be of the momentary-contact type.

(d) The station POWER-ON indicator light (green) shall be wired to the ON-OFF switch of the motor controller.

(e) The READY-TO-RUN pushbutton switch shall be of the momentary-contact type, wired to the READY-TO-RUN indicator light (white), and shall be interlocked with the CYCLE switch control circuit.

(f) Control station exposed to the weather shall have watertight enclosures, otherwise the enclosure shall be dripproof.

3.7.1.2 Labels. On each control stations, each indicator light and switch shall have a label to identify its function. In addition, the two-position switch (UP and DOWN) on the master control station shall have labels to indicate the UP position and the DOWN position.

3.8 Safety devices.

3.8.1 Overtravel limit safety device. The conveyor shall have an overtravel limit device installed at the top and bottom load stations. The overtravel limit device shall interlock with the control circuit to stop the conveyor if the palletized load on the tray is carried past the top or

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bottom load station. The control circuitry shall permit conveyor operation in the direction to release the pallet load (see 4.6.5.1).

3.8.2 Load-out-of-position safety device. The conveyor shall have a load-out-of-position safety device installed in the trunk at the top and bottom of each trunk door opening in line with the vertical face of the trunk liner. Each load-out-of-position device shall be a photo-electric eye system able to detect a load that extends past the edge of the tray. The devices shall be interlocked with the conveyor control circuitry to prevent any movement of the conveyor. Each photo-electric eye system, when cleared of the displaced load, shall reset to provide the protective circuitry for load-out-of-position protection and to permit operation of the conveyor (see 4.6.5.2).

3.8.3 Electrical overload device. The drive unit shall be equipped with an electrical overload device to monitor the motor output power with meter readout which will open the motor circuit in the event of conveyor overload of 115 percent of the conveyor rated capacity. The device shall be in addition to the motor thermal overload relay incorporated in a motor controller. It shall be adjustable to permit a setting accuracy (plus or minus 5 percent maximum) and shall by-pass the power surge of current and torque normally associated with starting the motor, but shall operate to open the motor circuit in the event of motor overloads. The device shall be adjustable to 150 percent of conveyor rated capacity . The device shall include means for reset of protective circuitry. The meter readout device shall be mounted on the motor controller enclosure door so that it may be viewed with the enclosure door closed (4.6.5.3).

3.8.4 Audible warning device. A sound signal device shall be provided at each load station. A time-delay relay of 3 to 3-1/2 seconds duration shall be provided in the conveyor control circuitry. The signal device shall be integrated into the conveyor control circuitry to energize the sound device, prior to movement of the conveyor, at any station with its trunk door open. The sound signal shall continue for the 3 to 3-1/2 seconds duration of the time-delay of the relay. The sound signal shall stop when the conveyor starts moving. The sound signal device shall be of the buzzer or vibratory horn type with an adjustable sound pressure level of 40 to 85 decibels A-weighted (dbA) (ref 20 micro-Pascals) (see 4.6.5.4).

3.8.5 Warning plates. A 5- by 9-inch warning plate shall be installed adjacent to each trunk door so as to be visible prior to and during conveyor operation. The warning plate shall be in accordance with MIL-P-15024, Type H, and MIL-P-15024/5, style I or IV, and inscribed as follows:

WARNING

KEEP CLEAR OF TRUNK

OPENING DURING

CONVEYOR OPERATION

The plate shall be visible in both low level red and white illuminated areas of a ship.

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3.8.6 EMERGENCY STOP switch. A momentary-contact type EMERGENCY STOP switch, with a red mushroom head, shall be provided at each control stations. In addition, an EMERGENCY STOP switch shall be provided in the conveyor machinery space. Depressing any of the EMERGENCY STOP pushbutton switches shall stop the conveyor and shall trip contactors in the electric motor controller. To restore power to the motor, the controller shall be reset before the conveyor can be restarted.

3.8.7 Trunk door interlock switch system. For conveyors serving three or more levels, a trunk door interlock switch system shall be provided that permits conveyor operation only when the door at the deck with the master control and one other trunk door is open. The opening of a third trunk door, shall prevent or stop the operation of the conveyor.

3.9 Identification plate. An identification plate shall be attached to a part of the conveyor which will not be renewed during its normal life. The plate shall be readily visible. The identification plate shall be in accordance with MIL-P-15024 and MIL-P-15024/5 and shall be metal with marking and engraving as specified for type B plates. Plate dimensions shall not exceed the dimensions shown for a number 6 size plate. Fasteners used to secure plate shall be corrosion-resistant.

3.9.1 Identification plate markings. The minimum information to be on the identification plate for the conveyor shall include the following items:

(a) Item name (vertical conveyor, tray type).

(b) Design characteristics - conveyor weight, tray capacity, conveyor rated capacity, motor horsepower, motor revolutions per minute, class shock (grade B).

(c) Specification number.

(d) Manufacturer's name, model, and serial number.

(e) Contract or order number.

3.10 Operating instruction chart. An operating instruction chart shall be provided at each control station of the conveyor. The chart shall include operating and safety instructions listed in figure 3, modifed to suit the specific shipboard installation. Charts shall be of laminated plastic material or of aluminum as used in the metal photo process in accordance with MIL-P-15024, type F or H and MIL-P-15024/5. The charts shall be installed in the immediate vicinity of each conveyor trunk opening as a visible reference for operating personnel.

3.11 Painting. Structural parts and exposed surfaces, except chains, cam tracks, and surfaces in rolling or sliding contact shall be cleaned and pre-treated. Surface cleaning shall be in accordance with the applicable method of TT-C-490. Pretreatment coating shall be in accordance with Type III of TT-C-490. The pre-treated surface shall be primed with either one coat of TT-

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P-645, 0.001 inch minimum dry film thickness or one coat of TT-P-1757 primer, 0.001 inch minimum dry film thickness. Finish coat shall be type I, II, or III, class 2, of MIL-E-15090, 0.001 inch minimum dry film thickness.

3.12 Noise. The noise level emanating from conveyor operations shall be not greater than 84 dbA when measured at any control station. (see 4.6.6)

3.13 Reliability. The conveyor shall operate for 250 cycles without failure. A cycle is defined in 4.6.9.1.

3.14 Maintainability. Conveyors shall be accessible for planned and corrective maintenance. Corrective maintenance requirements shall be accomplished by replacement of parts by ship's personnel. Corrective and planned maintenance tasks shall be not greater than the following values:

(a) Only one task of the 20 simulated planned maintenance tasks may be greater than 2 hours to accomplish, and no planned maintenance tasks shall be greater than 4 hours to complete.

(b) Not greater than two of the 20 corrective tasks shall be greater than 2 hours to accomplish, and the mean-time-to-repair (MTTR) shall not be greater than 3 hours.

Maintainability tasks shall be performed in accordance with test method 10 of MIL-STD-471 (see 4.6.11).

3.15 Communication system. A two-way sound powered telephone system between conveyor load stations shall be provided. The call signal stations shall be in accordance with MIL-C-15306 and one station shall be located above each conveyor control station. Each call signal station shall have a hard-wired handset and a jack to accommodate a portable head-chest set (see 4.6.7). The telephone handsets shall be in accordance with type H-203/U of MIL-T-15514. Two head-chest sets in accordance with type H-200/U of MIL-T-15514 shall be provided. The communication system shall not be integrated with any other shipboard circuits, but shall be solely for conveyor use.

3.16 Conveyor lighting system. At the highest level, fluorescent lighting fixtures shall be provided and located in the overhead of the trunk to illuminate the highest trunk opening and the tray at that level. At all other trunk doors, two fluorescent lighting fixtures shall be provided to illuminate the trunk. These fixtures shall be located in vertical openings in the front side liners and shall not interfere with conveyor operations. In addition, fluorescent lighting fixtures shall be provided for the drive machinery. All lighting fixtures shall provide both white and red (night) light. A switch, located near the master control station, shall energize either the white lights or the red lights (Note: In no case, shall it be possible to energize both the white and red lights simultaneously). The lighting system shall be independent of the conveyor circuitry.

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3.17 Elapsed time meter. An elapsed time meter, of the non-resettable type, in accordance with MIL-M-7793 and MIL-M-7793/8, graduated in hours up to 9999 hours shall be integrated in the conveyor drive motor electrical circuit to monitor the actual operating time of the conveyor. The elapsed time meter shall be mounted on the motor controller enclosure door so that the meter may be viewed with the enclosure door closed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements .

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. The first conveyor ordered under a contract or purchase order shall be subjected to the examinations listed in 4.5 and the tests specified in 4.6. For a repeat design conveyor, a certificate of compliance shall be prepared for evidence of accomplishment of the tests and demonstrations on 4.6.8, 4.6.9, and 4.6.11 (see 6.3). If any component in the conveyor is changed, then it is not considered to be a repeat design.

4.4 Quality conformance inspection. Each conveyor shall be subjected to the examinations listed in 4.5 and the tests specified in 4.6 except for 4.6.8, 4.6.9, and 4.6.11.

4.5 Examination. Each conveyor shall be visually examined for defects listed in table II. Presence of one or more defects shall be cause for rejection.

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TABLE II. *Visual defects.*

Defect	Requirement
Use of cast iron parts.	3.1.1
Cam track, material not as specified.	3.1.2
Roller chain and attachments, material and design not as specified.	3.1.3
Chain sprockets, design not as specified.	3.1.4
Keys, collars, caps, fasteners, and other parts not positively fastened.	3.2.2
Lubrication fittings not provided, and access to machinery not as specified.	3.2.3
Bearings not as specified.	3.2.4
Liners not provided or not as specified.	3.2.5
Tray carrier chain tensioner not provided.	3.2.6
Cam tracks do not provide positive control and guidance of tray travel; head and tail cam track panels not of fixed center design specified; track section fastening not as specified.	3.2.7
Access to conveyor pit not provided.	3.2.8
Conveyor frame not equipped with lifting attachments for handling; Conveyor trunk not designed to withstand pressure due to watertight requirements.	3.2.9
Conveyor configuration not as specified.	3.2.10
Number of trays and tray spacing not as specified.	3.2.10.1
Drive machinery arrangement not as specified.	3.2.10.2
Number of conveyor length sections not as specified.	3.2.10.3
Welded fabrication, welder qualification, and inspections not as specified.	3.2.10.4
Design does not include loading effect due to ship motion.	3.2.11
Drive machinery, mounting and location differs from approved shock tested unit.	3.2.12
Design criteria for strength of conveyor not as specified.	3.3
Stress relationship for stress analysis not as specified.	3.3.1
Drive and tray carrier chain, safety factor and chain construction not as specified.	3.3.2
Conveyor support from ship structure not as specified.	3.4
Tray attachment and support not as specified.	3.4.1
Conveyor load area less than minimum dimensions specified. Conveyor cross section greater than maximum dimensions specified.	3.4.2
Trunk doors and openings not as specified.	3.4.3
Conveyor operating cycle not as specified.	3.5
Tray capacity, conveyor rated capacity, and speed not as specified.	3.5.1
Drive machinery not as specified.	3.6.1
Electric drive motor not as specified.	3.6.1.1
Electric motor controller not as specified.	3.6.1.2
Magnetic clutch not as specified, certification missing.	3.6.1.3
Speed reducer not as specified.	3.6.1.4
Speed reducer casing not as specified.	3.6.1.4
Brake not as specified.	3.6.1.5
Machinery bedplate not as specified.	3.6.1.6

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TABLE II. Visual defects (continued).

Defect	Requirement
Chain tensioner missing, location not as specified.	3.6.2.1
Chain sprocket assembly not as specified.	3.6.2.2
Shaft mounting not as specified. For welded shafts, required inspections (X-ray, fabrication) not performed or not as specified.	3.6.2.3
Conveyor controls do not function as specified.	3.7
Control station and devices not as specified.	3.7.1
Control station controls not as specified.	3.7.1.1
Labels not as specified.	3.7.1.2
Overtravel limit safety device not as specified.	3.8.1
Load-out-of-position safety device not as specified.	3.8.2
Electrical overload device not as specified.	3.8.3
Audible warning device not as specified.	3.8.4
Warning plates missing or not as specified.	3.8.5
EMERGENCY STOP switches missing or not as specified.	3.8.6
Trunk door interlock switch system not as specified.	3.8.7
Identification plate missing or not as specified.	3.9
Identification plate marking not as specified.	3.9.1
Operating instruction chart missing or not as specified.	3.10
Painting, surface preparation or paint coating not as specified.	3.11
Communications systems not as specified.	3.15
Conveyor lighting system not as specified.	3.16
Elapse time meter not as specified.	3.17

4.6 Tests. Each conveyor shall be subjected to the tests specified in 4.6.1 through 4.6.7 after installation in the ship. Tests shall be conducted with test package, size and weight as specified. Failure ofany test shall be cause for rejection of the conveyor. Prior to conducting any test requiring a post-test inspection to determine its success or failure, a thorough visual inspection shall be performed to establish a reference base for the post-test inspection.

4.6.1 Static overload. The conveyor shall demonstrate ability to hold total static load equal to 200 percent of the tray capacity multiplied by the number of trays in the horizontal position for a period of 10 minutes. Evidence of damage or permanent deformation of any component shall be cause for failure of the test.

4.6.2 Dynamic overload. The magnetic clutch, brake, and electrical overload device shall be adjusted to permit operation of conveyor in overload mode. The conveyor shall be loaded to 150 percent of the conveyor rated capacity (see 3.5.1) and operated for one cycle as defined in 4.6.9.1. Under this test condition, there shall be no overheating of drive machinery (see 3.6.1); there shall be no evidence of binding of moving parts; there shall be no distortion of equipment or structure; and there shall be no noises which could be indicative of equipment malfunction or component overstress. Failure of conveyor to operate or evidence of conveyor malfunction shall be cause for failure of the test. After completion of the overload test, the magnetic clutch and brake shall be readjusted for conveyor rated capacity and the electrical overload device adjusted to provide conveyor overload protection (see 3.6.1, 3.6.1.5 and 3.8.3).

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4.6.3 Conveyor rated capacity. The conveyor shall be cycled continuously for not less than 2 hours. Conveyor operation shall consist of alternate lowering and hoisting cycles. Each cycle shall start with all conveyor trays empty, proceed by loading a 3000 pound palletized test load onto each of 15 successive trays, and terminate when the 15 test loads have been transferred from the loading station to the unloading station. The top and bottom stations shall be used for this test. Maximum elapsed time between cycles shall be not greater than 2 minutes. A loading time and unloading time of 15 seconds each will be allowed per palletized load. Under this test condition, there shall be no overheating of drive machinery (see 3.6.1); there shall be no evidence of binding of moving parts; there shall be no distortion of equipment or structure; and there shall be no noises which could be indicative of equipment malfunction or component overstress. Stopping of conveyor for maintenance or repair shall be cause for failure of the test.

4.6.4 Non-overhauling speed reducer. The conveyor shall be loaded to the conveyor rated capacity (see 3.5.1). With the conveyor stopped, the electric brake shall be manually released. Rotation of the speed reducer input shaft due to loads on the trays shall be cause for failure of the test.

4.6.5 Safety devices.

4.6.5.1 Overtravel limit device. A test load (45 inches deep by 54 inches wide by 18 inches high and not greater than 300 pounds) shall be placed on a conveyor tray and the conveyor shall be operated in the DOWN direction. The overtravel limit device at the bottom load station shall stop the conveyor before the loaded tray rotates or jams in the idle tray return cycle. Successful operation of the stopped conveyor in the UP direction and removal of the test load through the trunk opening shall be demonstrated. The conveyor, with test load, shall be operated in the UP direction so that the moving load impinges on the overtravel limit device at the top load station. The overtravel limit device shall stop the conveyor before the loaded tray rotates or jams in the idle tray return cycle. Successful operation of the stopped conveyor in the DOWN direction and removal of the test load through the trunk opening shall be demonstrated. Failure to limit overtravel, failure to operate to release test load, or evidence of damage to test load or conveyor shall be cause for failure of the test.

4.6.5.2 Load-out-of-position safety device. Each load-out-of-position safety device shall be tested. A test load (45 inches deep by 54 inches wide by 18 inches high and not greater than 300 pounds) shall be placed on a tray so that it extends over the front edge of the tray past the vertical face of the conveyor front package liners. Attempt to operate the conveyor in both directions to verify that the photo-electric eye light beams have been interrupted by the test load and that the conveyor will not move. With the test load removed, the photo-electric eye unit shall reset automatically to provide the protective jam safety circuit. Failure of the photo-electric eye unit to function to prevent the conveyor from operating shall be cause for failure of the test.

4.6.5.3 Electrical overload device. The conveyor shall be operated in the UP direction at conveyor rated capacity, using the bottom load station to load the test loads. With the conveyor operating at conveyor rated capacity, electrical readings of the motor line voltage and amperage shall be recorded. The conveyor shall be stopped, and the test loads returned to the bottom load station. With test loads equal to 115 percent of tray capacity, the conveyor shall be operated in

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the UP direction using the bottom load station to load the conveyor. Electrical readings of the motor line voltage and amperage shall be recorded when the electrical overload device trips out the motor circuit. Failure of the electrical overload device to trip out the motor circuit at 115 percent of conveyor rated capacity shall be cause for rejection. After completion of the test, the electrical overload device shall be reset to provide conveyor overload protection, and the conveyor shall be satisfactorily operated at conveyor rated capacity.

4.6.5.4 Audible warning device. At each load station, with the trunk door open, the conveyor shall be operated in the UP and DOWN directions. The audible signal shall sound when the conveyor CYCLE pushbutton is actuated. The sound signal shall continue for the 3 to 3-1/2 seconds duration of the time-delay of the relay. The sound signal shall stop when the conveyor starts. It shall be demonstrated that the sound level of the signal device is adjustable over its full range (see 3.8.4). The sound pressure level of the audible warning device shall be measured at a distance of 3 feet using a sound level meter in accordance with ANSI S1.4, type 2. Failure of the audible signal device to function as required shall be cause for failure of the test.

4.6.5.5 Trunk door interlock switch system. For conveyors serving three or more levels, the following shall be demonstrated on all combinations of trunk doors:

(a) Open the trunk door at deck with the master control station and any other trunk door and operate the conveyor in both directions.

(b) Operate the conveyor in down direction, open a third door, observe that the conveyor stops, close the third door, and cycle the trays to the next level. Repeat this demonstration with the conveyor operating in the up direction. Continued movement of the conveyor in either direction shall be cause for rejection.

(c) Open three trunk doors and attempt to operate the conveyor in both directions. Any movement of the conveyor in either direction shall be cause for rejection.

4.6.6 Noise test. The conveyor shall be noise tested under normal operating conditions in accordance with MIL-STD-740-1 and MIL-STD-740-2 . Measurements shall be made at all control stations to ensure that the requirements of 3.12 are met.

4.6.7 Control stations. The control station at each load station and machinery space shall be tested. With electrical power to the conveyor available, the following shall be demonstrated:

(a) With control station secured (lockout device in place):

(1) The STOP-ON switch shall be set to ON position. The green indicator light (POWER ON) shall be visible.

(2) The STOP-ON switch shall be set to STOP position, the green indicator light (POWER ON) shall be out and the red indicator light in the STOP-ON switch shall be visible.

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(3) Demonstrate the master control station UP-DOWN selector switch and the CYCLE pushbutton switch cannot be actuated and conveyor motion cannot be started with load station STOP-ON switch set to the STOP position.

(4) The conveyor shall be run from the master control station, the conveyor and motor shall be stopped using the STOP switch at a secured station.

(5) The lockout device is adequate to secure the master control station against unauthorized conveyor operation.

(6) The communication system is operating at all control stations using both hard-wired handsets and portable head-chest sets.

(b) With the control station open (lockout device removed):

(1) The conveyor shall be operated in the UP and DOWN direction and the following shall be demonstrated:

a. The conveyor will move the distance between trays and stop automatically at load stations with trays in position for simultaneous load-unload operation.

b. The automatic stop (cycle) of the conveyor is accomplished by de-energizing the magnetic clutch and setting the brake. The drive motor is not stopped (the white indicator light MOTOR RUNNING should be visible).

c. Interrupt the photo-electric eye light beam, the red indicator light LOAD-OUT-OF-POSITION should show at the master control station and at the load station that the light beam is broken.

d. Actuate the READY-TO-RUN pushbutton switch at the load stations, the white indicator light, at the master control station should be visible for the load stations reporting.

e. At the master control station, demonstrate that the conveyor, with motor running, can not be cycled if the READY-TO-RUN circuits have not been actuated.

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f. It shall be demonstrated that the conveyor cannot be operated unless the STOP-ON switch is set to ON position at all control stations.

Failure of the control stations to function as described shall be cause for failure of the test.

4.6.8 High-impact shock test. The conveyor drive machinery shall be shock tested in accordance with MIL-S-901 (see 6.3). The conveyor drive shall be secured to a test fixture attached to the anvil table of the shock machine. A conveyor that has a protective enclosure for the conveyor drive machinery, both the conveyor drive machinery and the protective enclosure shall be secured to the test fixture, in the same manner and attitude as installed on the conveyor for the shock test. Parts coming adrift or not contained within the protective enclosure (if provided) shall be cause for failure of the shock test.

4.6.9 Reliability demonstration. A reliability test program shall be conducted on the conveyor system after being installed on board a ship. A failure shall be as specified in 4.6.10.

4.6.9.1 Cycle test procedures. The conveyor shall be operated for 250 cycles without failure. After every 50 cycles, the conveyor components shall be inspected for evidence of damage or deformation. The test loads for all reliability cycle and jam tests shall be 3000 pounds and shall be 45 inches deep by 54 inches wide by 56 inches high on 40 x 48 inch pallets. The conveyor shall be cycled for not less than 10 hours a day. A cycle is defined as follows:

(a) Each cycle shall start with all the trays empty.

(b) The tray at the highest level shall be designated as the "first tray".

(c) Load a test load on the "first tray" and operate the conveyor in the DOWN direction until the next tray is in position to load.

(d) Continue to load the trays and operate the conveyor in the DOWN direction until the "first tray" reaches the lowest level. Then, unload the test load from that tray. Continue to operate the conveyor in the DOWN direction, loading (at the highest level) and unloading (at the lowest level) test loads on the trays until the "first tray" returns to the highest level.

(e) With the "first tray" empty and the other trays (in horizontal orientation) still loaded, operate the conveyor in the UP direction, loading (at the lowest level) and unloading (at the highest level) test loads on the trays until all the test loads have been returned to the highest level (The "first tray should be at the highest level and empty).

4.6.9.2 Jam test procedures. During the first 64 cycles of the reliability test, the conveyor shall be jammed deliberately during every other cycle. All jams in the "UP" direction shall be simulated by having the test load to be jammed, positioned so that it rests on the tray but extends through the trunk door. All jams in the "DOWN" direction shall be simulated by having the test load to be jammed, positioned so that it rest on the trunk door sill, but extends into the trunk.

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Further, all the remaining trays (in the horizontal orientation) shall be loaded with a test load. For all the jam tests, the load-out-of-position device (see 3.8.2) shall be disconnected and jumpered and the magnetic clutch (see 3.6.1.3) set to allow a force against the tray equal to 150 percent of conveyor rated capacity load. Jams shall be in accordance with the following:

(a) The first 20 jam tests shall be accomplished as follows:

(1) With the test load to be jammed, positioned on the tray at the highest level, the conveyor shall be jammed 5 times during a strike-up operation and then, 5 times during a strike-down operation.

(2) With the test load to be jammed, positioned on the tray at the lowest level, the conveyor shall be jammed 5 times during a strike-up operation and then, 5 times during a strike-down operation.

(b) The last 12 jam tests shall be accomplished with the instantaneous electrical overload devices (see 3.8.3) disconnected and jumpered and as follows:

(1) With the test load to be jammed, position on the tray at the highest level, the conveyor shall be jammed 3 times during a strike-up operation and then, 3 times during a strike-down operation.

(2) With the test load to be jammed, position on the tray at the lowest level, the conveyor shall be jammed 3 times during a strike-up operation and then, 3 times during a strike-down operation.

4.6.10 Criteria. Failure of the conveyor system shall be defined as any of the following events:

(a) Failure of the conveyor to perform in accordance with performance requirements of this specification (see 3.5).
(b) Permanent deformation of any load bearing or structural part of the conveyor as a result of normal operation or which occurs as a result of load jam.

(c) Failure of the conveyor to be returned to operation within 15 minutes after the occurrence of a load jam.

4.6.11 Maintainability demonstration. A maintainability demonstration test program shall be conducted on the conveyor system in accordance with the requirements of 3.14. The maintenance tasks to be performed shall consist of the replacement of parts listed in table III in accordance with test method 10 of MIL-STD-471. A minimum of 10 maintenance tasks to be demonstrated will be selected by NAVSEA or the contracting activity. Corrective maintenance tasks may be simulated as an alternative to deliberately introducing defective components into the system. MTTR shall be determined through a simple average of total corrective maintenance time divided by the total number of maintenance actions.

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TABLE III. *List of replacement parts for maintainability demonstration.*

1. Switch proximity	12. Switch, selector, two positions
2. Relay, control	13. Lining, friction, electric brake
3. Relay, timing	14. Cell, photo-electric
4. Relay, double timing	15. Reflector, photo-electric
5. Transformer, control	16. Relay, time delay, photo-electric
6. Contractor, reverse	17. Spring, helical, compression, electric brake
7. Relay, instantaneous overload	18. Spring, electric brake
8. Coil, set thermal overload	19. Seal, electric brake
9. Fuse, control circuit	20. Coil, Electric brake
10. Assembly, control section	
11. Switch, limit	

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of reference documents listed in section 2, see 6.8.)

5.1 Preservation, packing, and marking. Conveyors, accessories, and software shall be preserved level A or C; packed level A, B, or C and marked in accordance with MIL-M-3184. The level of preservation and packing shall be as specified (see 6.2). Detached accessories shall be preserved and marked to the levels specified for the conveyor.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Vertical conveyors are intended for conveying palletized loads between decks on naval ships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.

(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

(c) Trunk tightness requirements (see 3.2.9 and 3.4.3).

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(d) When configuration dimensions are other than as specified (see 3.2.10).

(e) Decks to be served (see 3.2.10).

(f) Spacing between trays (see 3.2.10.1).

(g) Machinery arrangement to be right-hand or left-hand drive (see 3.2.10.2).

(h) When conveyor is to meet high-impact shock requirements (see 3.2.12).

(i) When the trunk door opening size is other than as specified (see 3.4.3).

(j) When tray capacity, conveyor rated capacity, and speed are other than as specified (see 3.5.1).

(k) Electric drive motor enclosure if other than dripproof (see 3.6.1.1(e)).

(l) Electric controller enclosure if other than dripproof (see 3.6.1.2(g)).

(m) Brake enclosure if other than dripproof (see 3.6.1.5(d)).

(n) Control station locations when other than as specified (see 3.7.1).

(o) Master control station deck (see 3.7.1.(a)).

(p) First article test required (see 4.3).

(q) Level of preservation and packing required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2	DI-DRPR-81000	Product drawings and associated lists	-
3.2 and appendix A	DI-GDRQ-80650	Design data and calculations	-

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3.2.4, 3.6.1.3(c), and 4.2	DI-MISC-80678	Certification/data report	10.3.1 does not apply
4.2	DI-NDTI-80809	Test/inspections reports	-
4.6.8	DI-ENVR-80708	Shock test report	-

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract and should be in accordance with MIL-M-15071.

6.5 Drawings. Drawings should be prepared as specified and forwarded to the acquisitioning activity for review of design. The drawings should contain the data necessary for the installation, operation maintenance, and repair of the electrical and mechanical aspects of the quipment.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 Spare and repair parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Certifications. Certification of specified components should be prepared as specified (see 6.3) and forwarded to the acquisistioning activity for review. The bearing load rating for all bearings used in the conveyor shall be certified by the bearing manufacturer. The magnetic clutch torque rating shall be certified by the clutch manufacturer for a minimum safety factor of 3 based on the full load torque requirement.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Subject term (key word) listing.

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Cam track
Conveyor rated capacity
Liners
Pushbutton station
Roller chain
Sprocket assembly
Tray
Tray capacity
Tray carrier chain
Tray guide arm
Trunk
Vertical track

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Review activity:
DLA-CS

Preparing activity:
Navy - SH
(Project 3910-N004)

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APPENDIX A

DESIGN DATA AND CALCULATIONS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included in the design data and calculations when specified in the contract or order. This appendix is mandatory only when data item description DI-GDRQ-80650 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DESIGN DATA AND CALCULATIONS TECHNICAL CONTENTS

30.1 Calculations. In addition to the requirements specified (see 3.2), the following shall be furnished:

(a) The conveyor center of gravity location, conveyor weight and stress analysis for conveyor trunk (when furnished to watertight requirements), carrier assemblies, and all other items where stress criteria have been set forth by this specification.

(b) Motor horsepower requirement (se 3.6.1.1).

(c) Brake and clutch torque requirement (see 3.6.1.5 and 3.6.1.3).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-23842B(SH)	2. DOCUMENT DATE (YYMMDD) 1993 JULY 14

3. DOCUMENT TITLE
CONVEYOR, VERTICAL, TRAY TYPE, FOR 3000 POUND PALLETIZED LOADS (SHIPBOARD USE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL ENGINEER: Mr. Steve Magdits, SEA 03W21	b. TELEPHONE (Include Area Code) (1) Commercial (703) 602-1846 (2) AUTOVON 332-1846
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 03Q42, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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